HNRNPH RTstop sites - G4 sensitive constructs

Mirko Brueggemann

June 25, 2025

Contents

1 Analysis Description

This analysis is based on the G4 peaks defined in the RTstop site report. We expect some of these peaks to change in their strength depending on the presence of NaCl or KCl. KCl is meant to stabilize G4 formation, therefore peaks should mostly go up when comparing to NaCl. Here we want to test which G4 peaks are sensitive for KCl and which are not.

Table 1: Merge and combine

Option	nRanges
inputRanges	46.326
mergeCrosslinkSites	42.271
minCrosslinks	42.271
minClSites	42.271
centerIsClSite	42.239
centerIsSummit	NA

2 Merge KCl and NaCl peaks

Initially peaks were called separately on each condition, KCl and NaCl. Here I combined both sets of peaks. To keep the desired width of 3nt I applied the merge and re-size routine on all peak midpoints. This essentially uses the combined coverage of all replicates in both conditions to decide on the center of any peak region that only partially overlaps in the two sets. In the end I annotate for each peak if it was seen in NaCl, KCl or both.

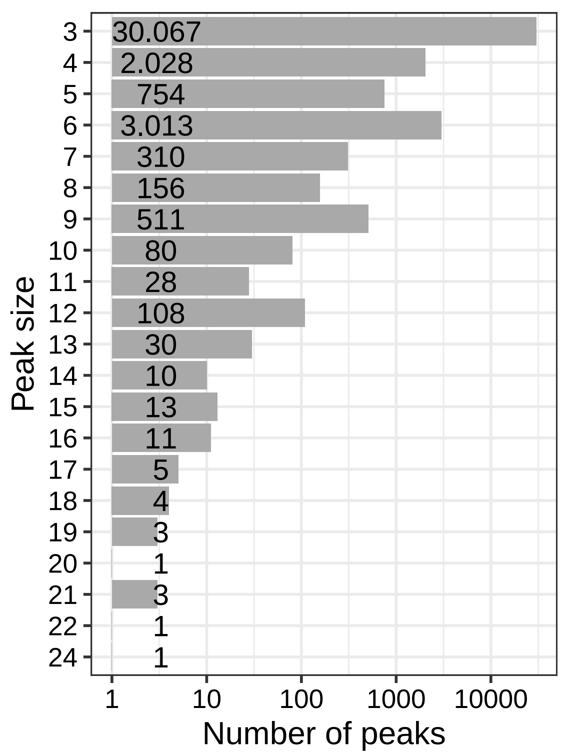

Figure 1: **Peak size distribution after merging of NaCl and KCl peaks.**

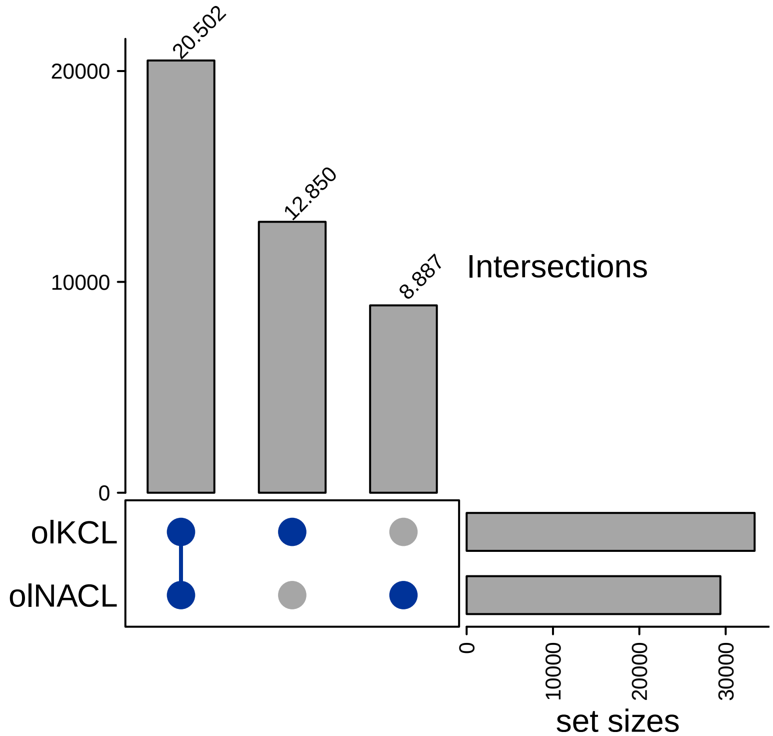

Figure 2: **Overlap between NaCl and KCl peaks.** Overlaps were computed after resolving partial overlaps.

3 Construct epxression

Constructs with in general low counts were removed from the differential analysis, because for these we don't have the count power to tell if they are sensitive to KCl or not. In order to do so, counts per construct were normalized to library size with DESeq2 and only constructs with at least 100 normalized counts from all 6 samples were retained.

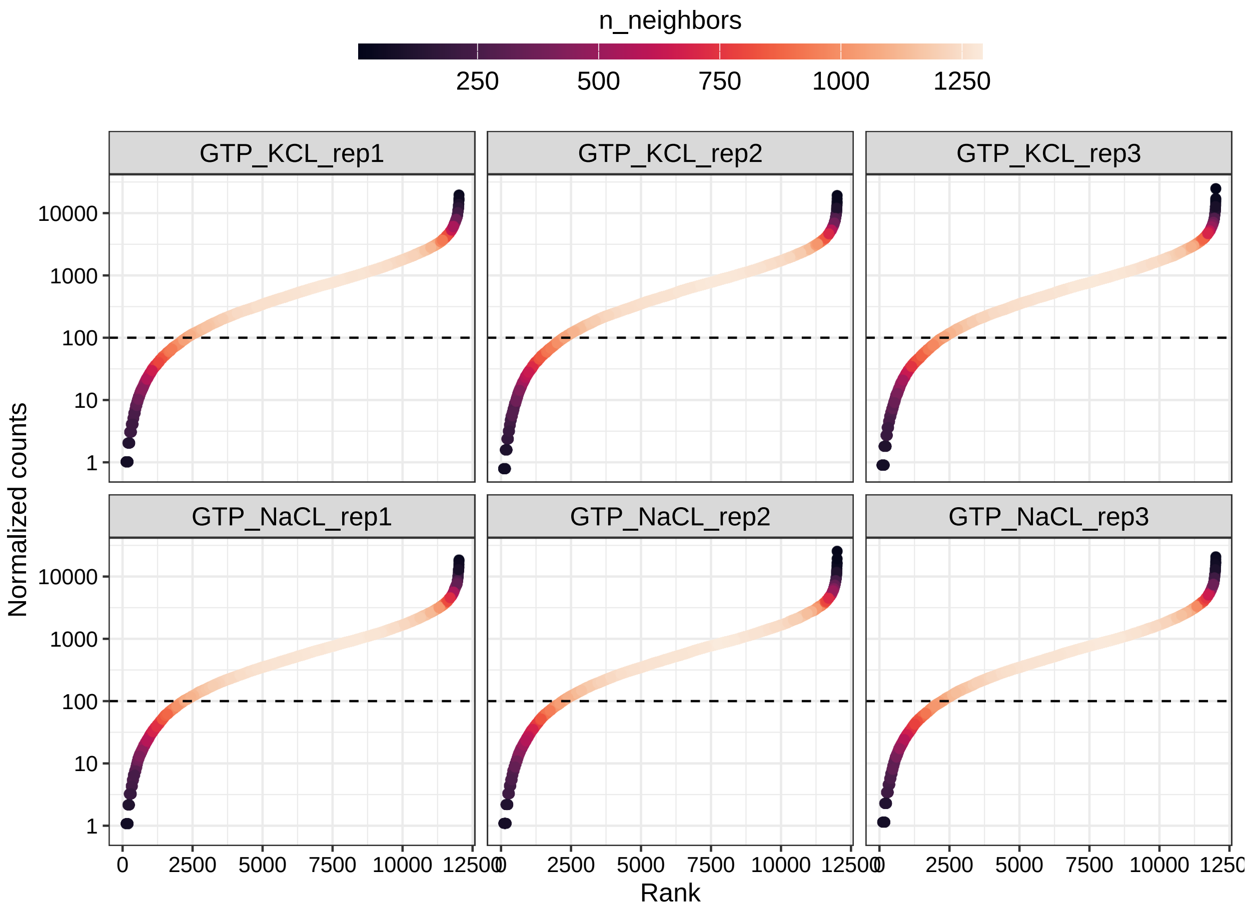

Figure 3: **Global expression.** Counts were normalized for library size with DESeq. These were summed up per construct and sorted by their rank. Only constructs with at least 100 normalized counts in all 6 samples are considered sufficiently large expressed.

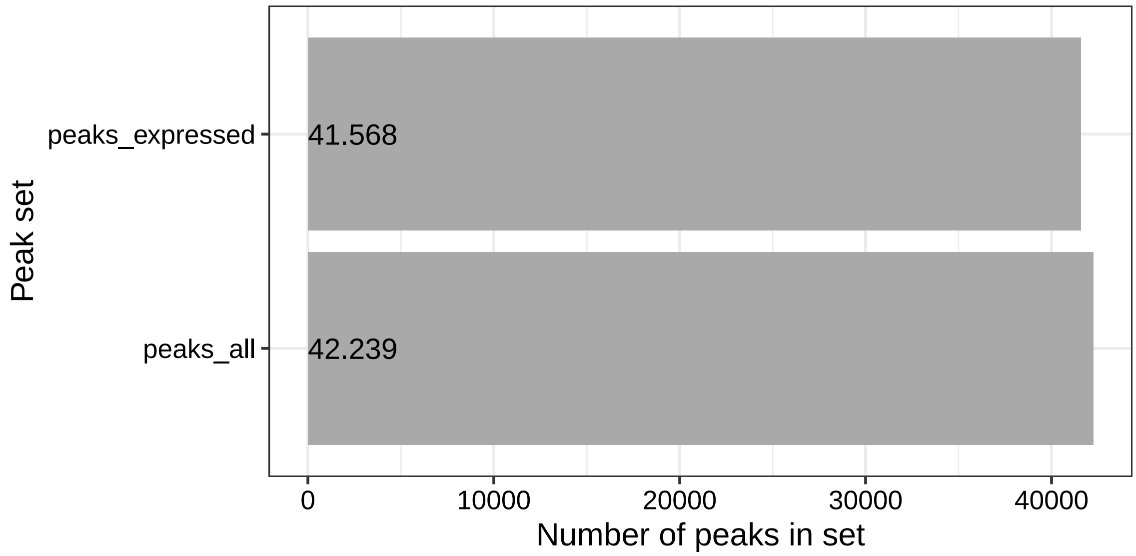

Figure 4: **Comparisson of all peaks and constructs vs. only those peaks on constructs considered expressed., eval=FALSE**

4 Differentially regulated peaks between NaCl and KCl

To compute fold-changes per peak for each construct between the NaCl and KCl condition I used a DESeq2 based approach. In order to remove the bias of constructs that strongly change in their expression between the conditions, I used a modified DESeq2 test. I specifically tested for those constructs that significantly change less than a log2 fold-change of 1 (less than 2-fold), using the paramteter altHypothesis = "lessAbs" and an lfcThreshold = 1 in the results function. This results in all constructs that are affected by a less than 2-fold global change between conditions. This was followed by a second round of DESeq2 on peak level for all peaks on constructs without major global changes. Here I now test for each peak if counts are different between NaCl and KCl (contrast = c("condition", "GTP_KCL", "GTP_NACL")).

NOTE:

- among the constructs that significantly changed globally about 80% of the individual peaks were also significant in the KCL condition
- These peaks are likely to be called significant, because of the underlying global transcript level change between KCL and NACL

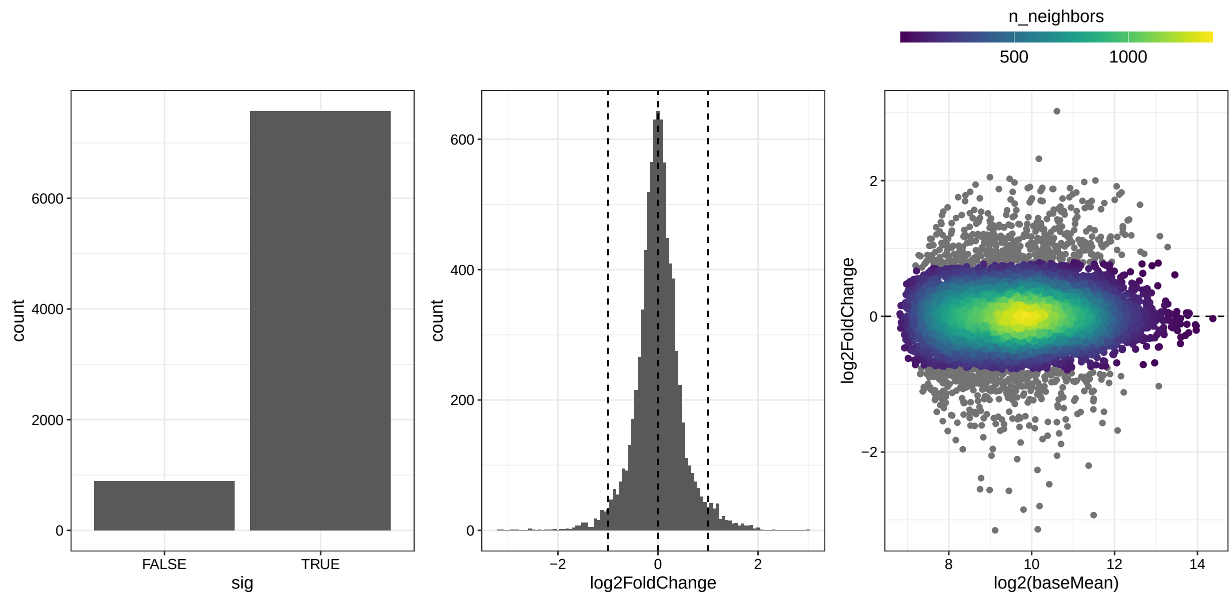

Figure 5: **Constructs that do not change more than 2-fold globaly.** Constructs are tested for an absolute global fold change to be less than 2-fold (LFC=1). Significant constructs do not change more than 2-fold on the total construct level.

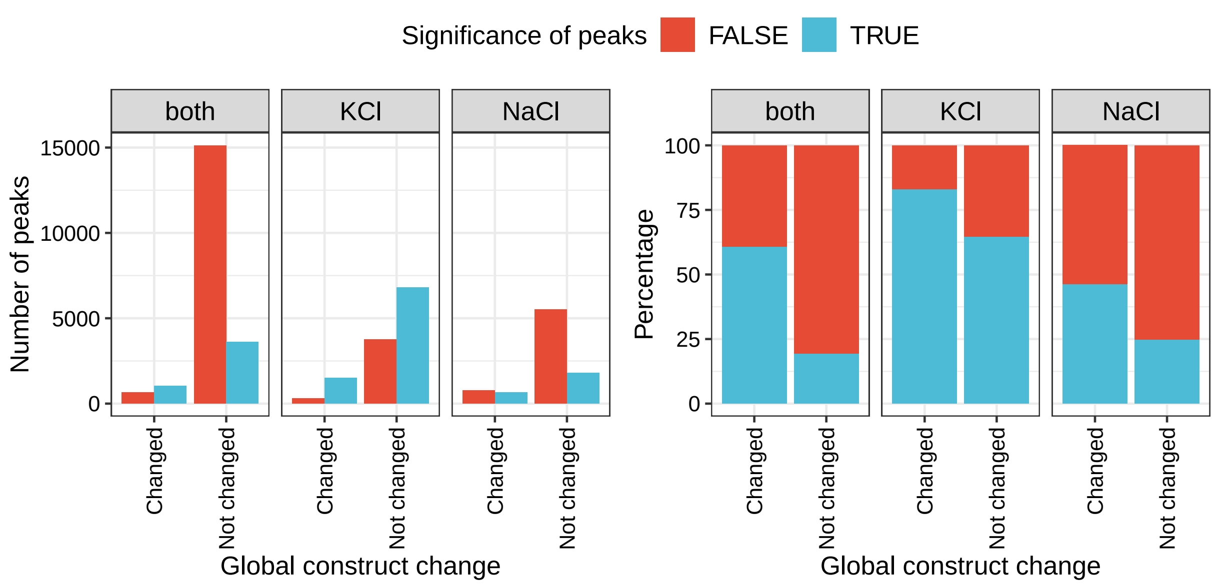

Figure 6: **Significant peaks on constructs that change/ do not change globally.**

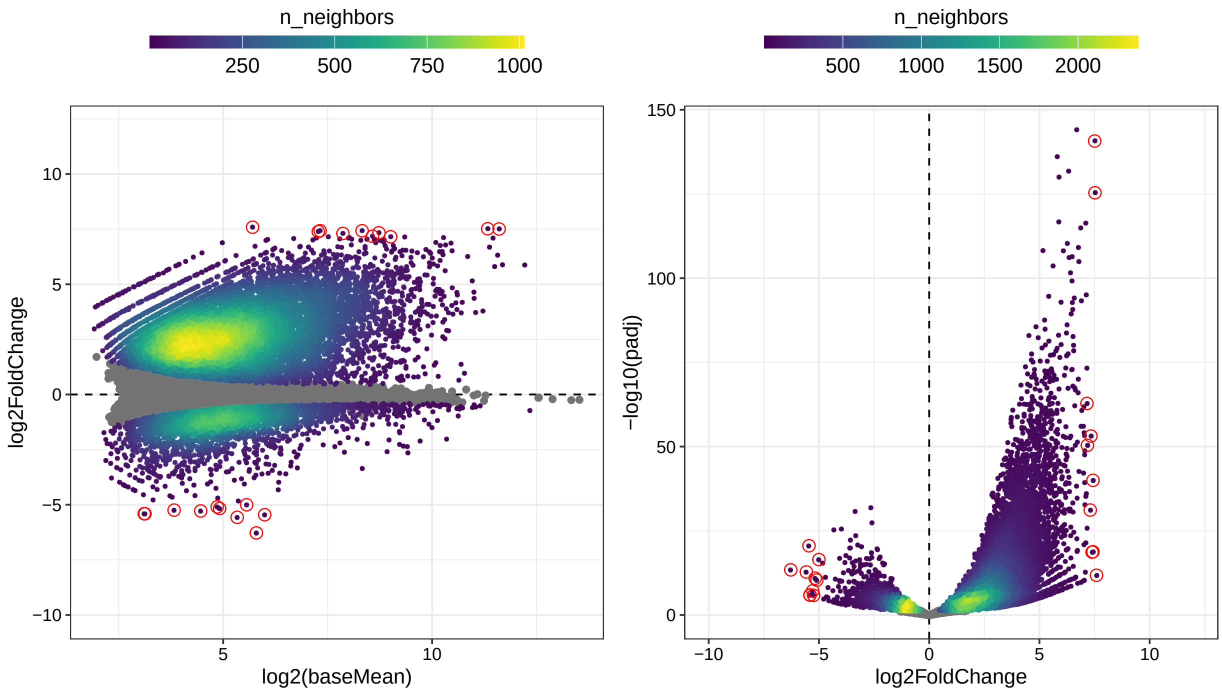

Figure 7: **Peak fold-changes without correction, on unchanged constructs.** Plots show the relationship of computed fold-change with the underlying expression strength (MA-plot) and the statistical significance in form of the adjusted P value (volcano-plot).

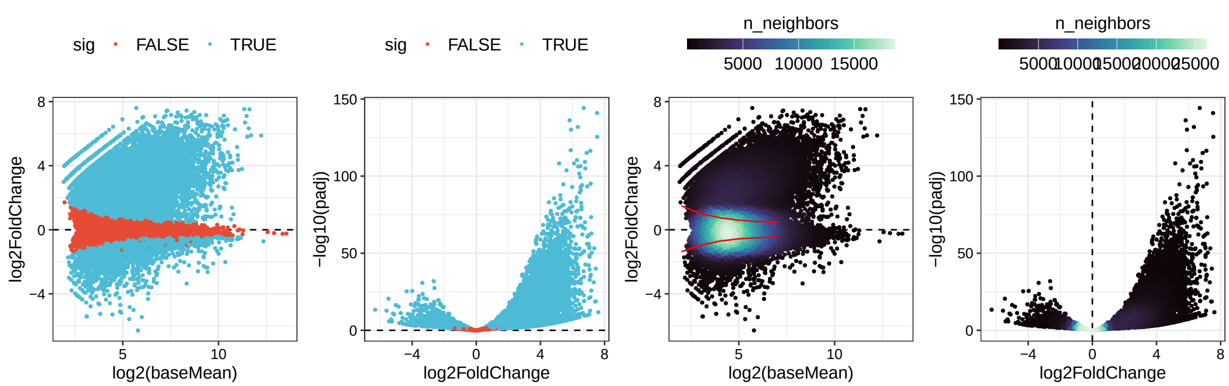

Figure 8: **MA and volcano variations**

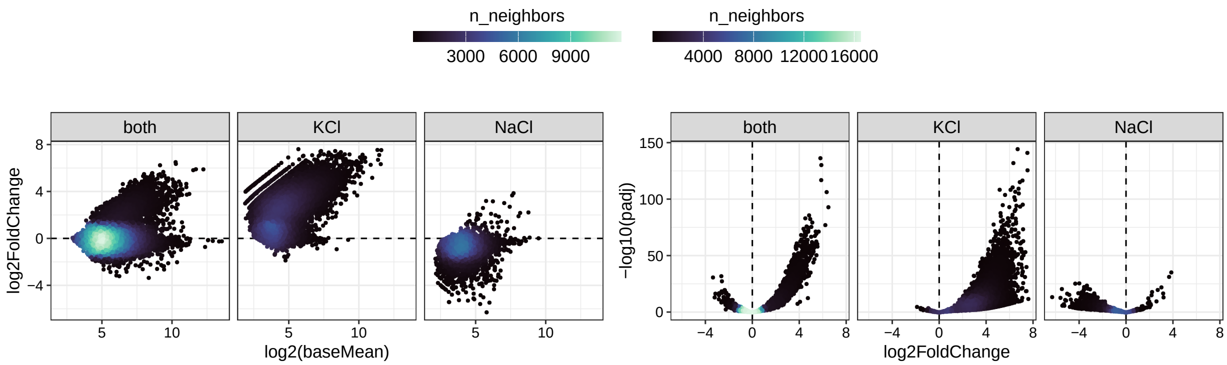

Figure 9: **Peak fold-changes without correction, on unchanged constructs split by peak support groups.** Plots show the relationship of computed fold-change with the underlying expression strength (MA-plot) and the statistical significance in form of the adjusted P value (volcano-plot).

5 Summary and numbers - tables

In summary this procedure yielded 12,217 significantly changed peaks on 4,532 constructs. From these 3576 constructs have at least one peak with the desired upwards shift. The total number of these peaks is 8,986

Table 2: Expression status of constructs.

Expression status	Value
0) All constructs	12,000
1) Not expressed	2,669
2) No Peak	868
3) Large global change	890
4) Final set	7,573

Table 3: Construct subclassification -old.

expressionStatus	sigStatus	dirStatus	highConf	n	per
Expressed + No Peak	-	-	-	868	7.23
Expressed + global change	-	-	-	890	7.42
Expressed + small change	FALSE	-	-	3041	25.34
Expressed + small change	TRUE	All down	-	956	7.97
Expressed + small change	TRUE	All up	FALSE	1725	14.37
Expressed + small change	TRUE	All up	TRUE	772	6.43
Expressed + small change	TRUE	Mixed	-	1079	8.99
Not expressed	-	-	-	2669	22.24

Table 4: Construct subclassification -new.

Type	Selection	#N	%
Expression	FALSE	2,669	22.2
Expression	TRUE	9,331	77.8
Peak	FALSE	890	9.5
Peak	TRUE	8,441	90.5
Global change	FALSE	868	10.3
Global change	TRUE	7,573	89.7
Significant	FALSE	3,041	40.2
Significant	TRUE	4,532	59.8
Direction	FALSE	956	21.1
Direction	TRUE	3,576	78.9

Table 5: Unique regions per construct. Number of constructs sums up to 12000. The number of regions does not sum up to 3000, because at this point a region can have constructs which were assigned to multiple categories. In these cases the regionID is counted multiple times.

status	nConst	nRegion	nPeaks	nPeaksSig	nPeaksSigUp
Not Expressed	2669	1315	NA	NA	NA
No Peak	868	527	NA	NA	NA
Global change	890	532	NA	NA	NA
Not Significant	3041	1314	11386	NA	NA
Significant-Down	956	578	5098	1524	NA
Significant-Up	3576	1306	20140	10693	8986

6 Summary and numbers - figures

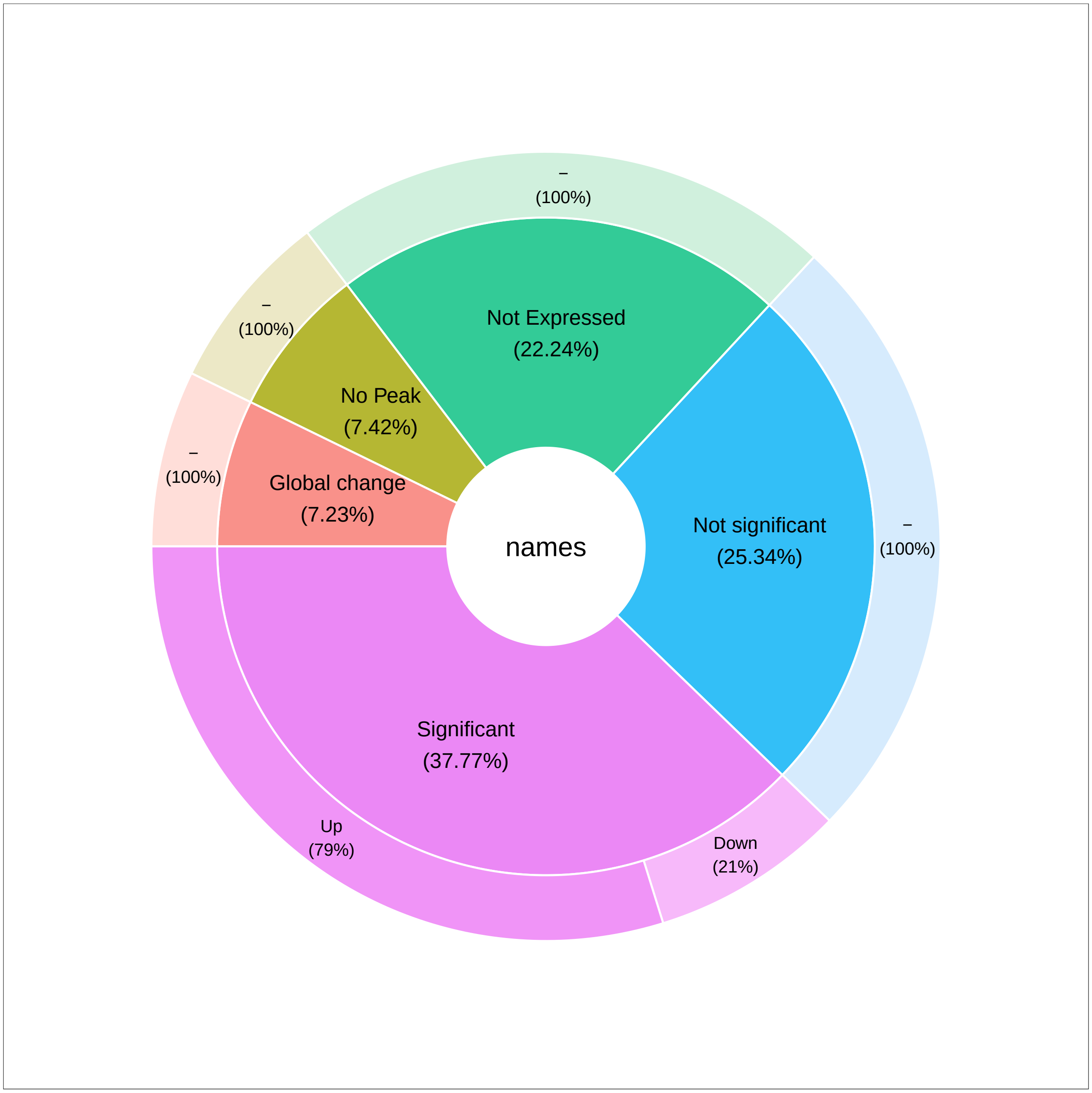

Figure 10: **Inner = Type of filtering step; Outer = The final set**

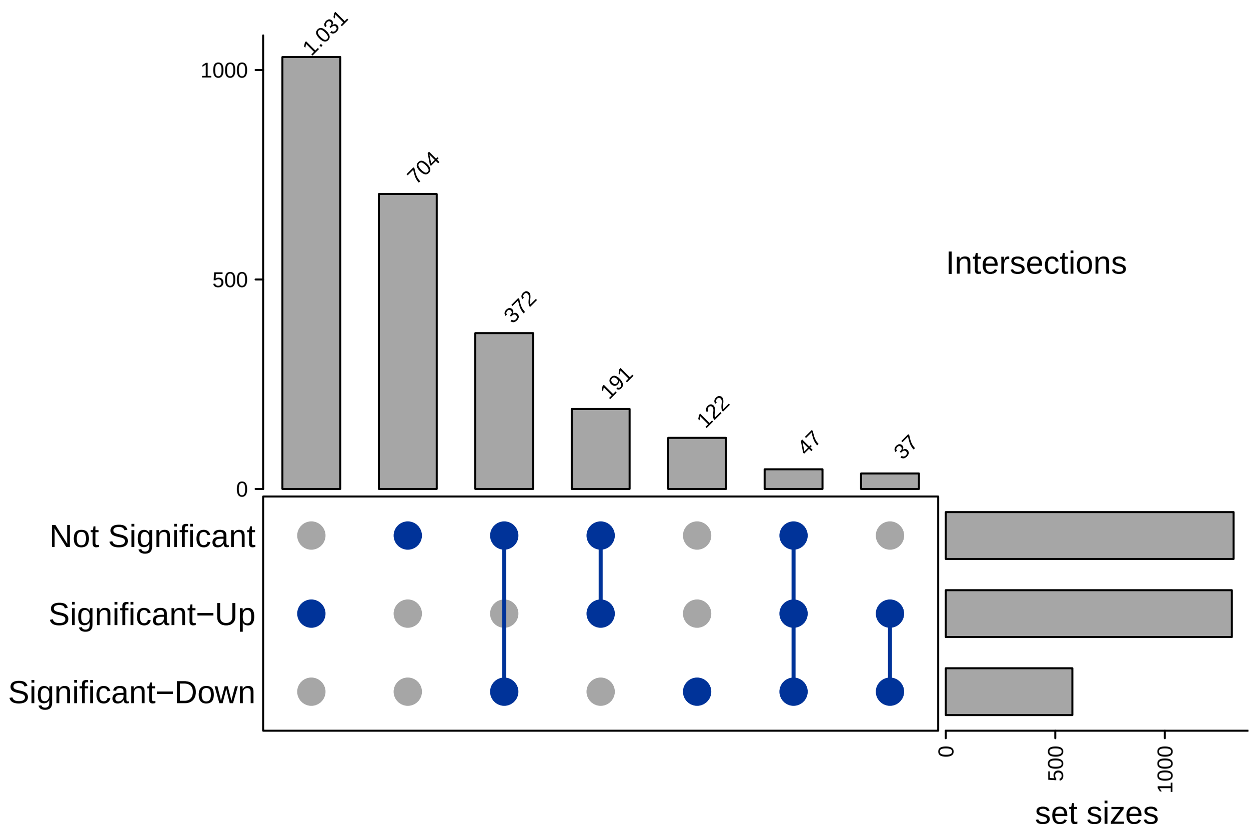

Figure 11: **Region distribution per classification category.** Each region can be assigned to multiple categories, based on the respective construct. Most regions (1.857) can be assigned to a single present category. Here all constructs were assigned to the same category (of the 3 presented); eg. for 1.031 regions all related constructs were assigned to the same category ('Significant-Up').

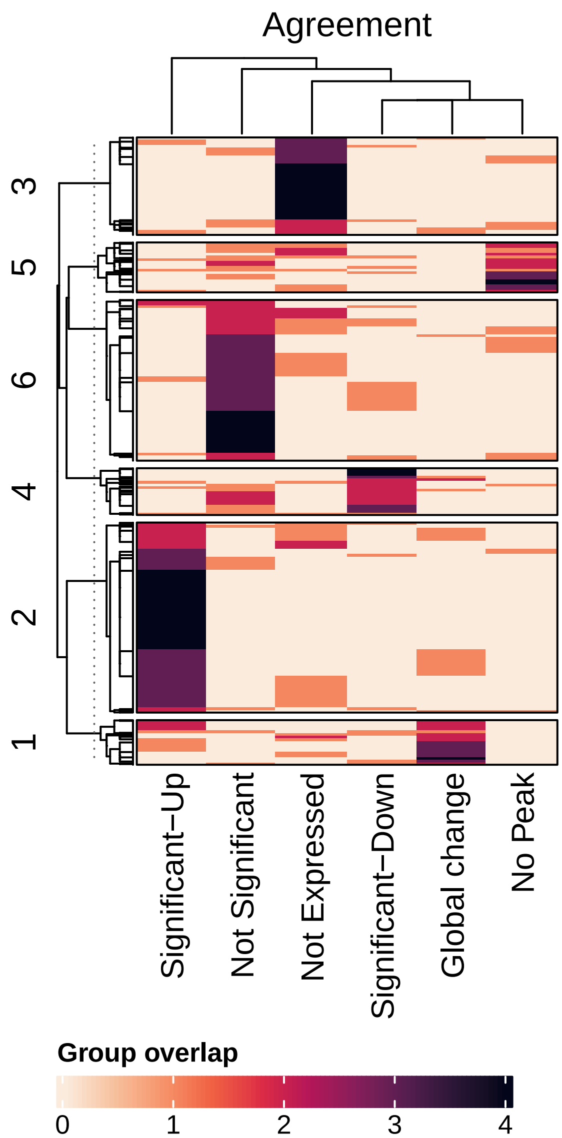

Figure 12: **Region and construct agreement based on classification type - For all constructs.** Rows = regions; cols = classification; n = number of constructs.

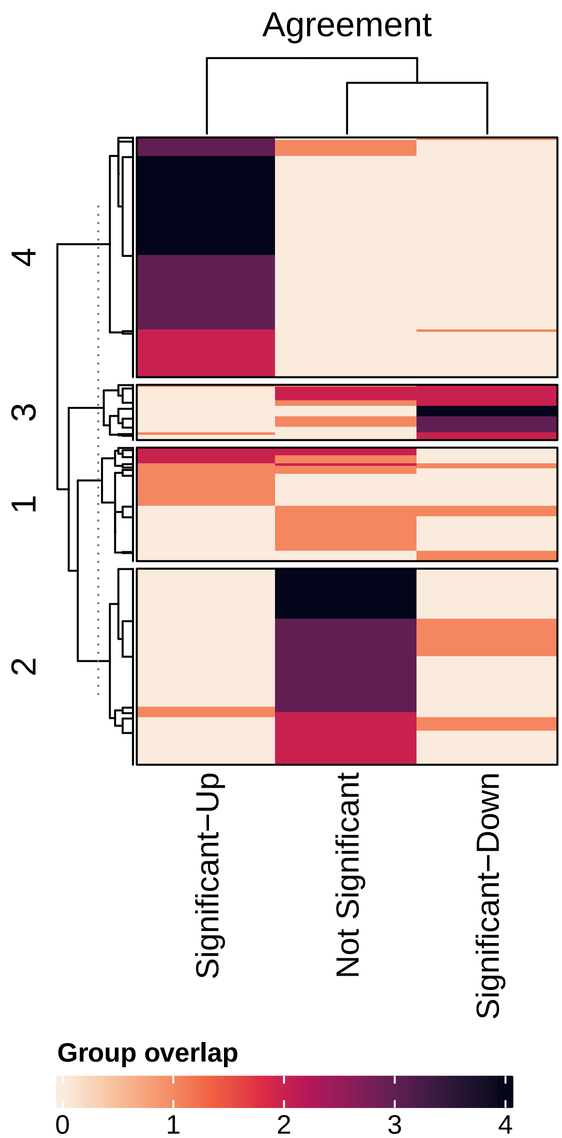

Figure 13: **Region and construct agreement based on classification type - Only for that survived filtering.** Rows = regions; cols = classification; n = number of constructs.

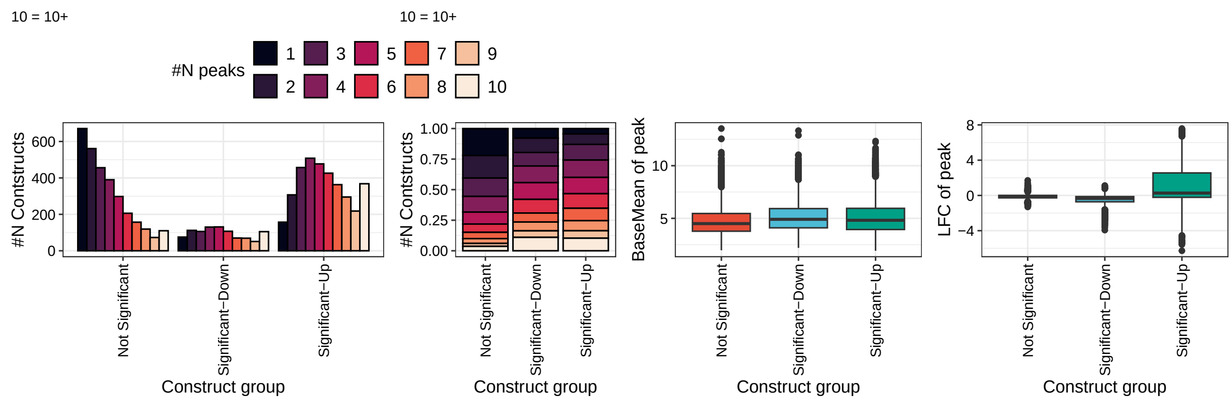

Figure 14: **Number of peaks per construct.** On all expressed peaks N=36.624.

7 G4 peak strength

G4 peak strength is calculated for each peak over two different sets. Set1 comprises of all peaks on G4 containing constructs, whereas set2 contains only the G4 peaks on the G4 containing constructs. The strength of each of these peaks is determined by dividing its counts by the total counts of the hosing construct.

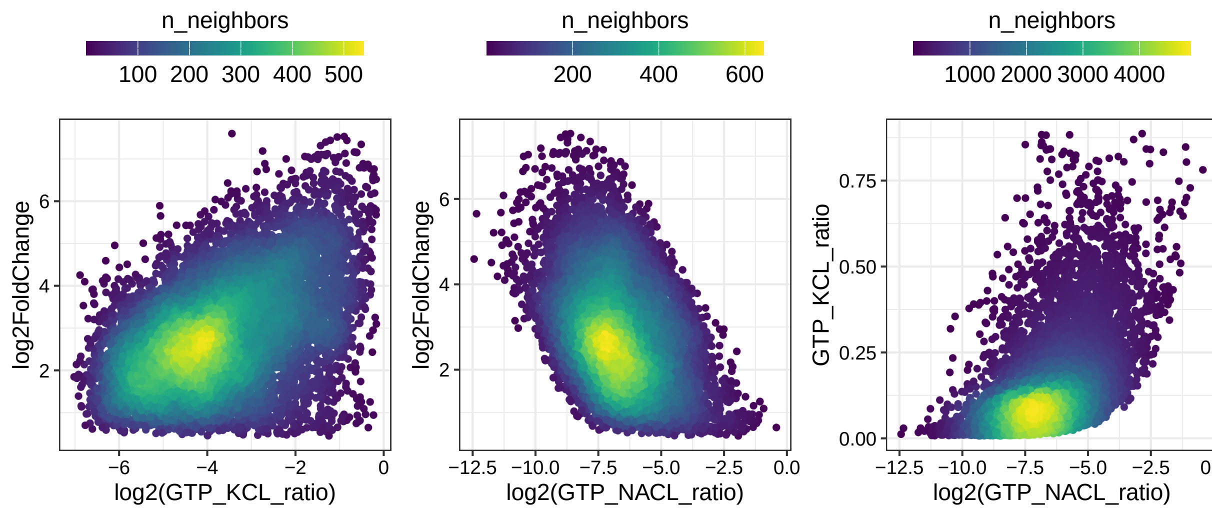

Figure 15: **Correlation between KCl/ NaCl peak ratios and the LFC.** G4 Peaks only.

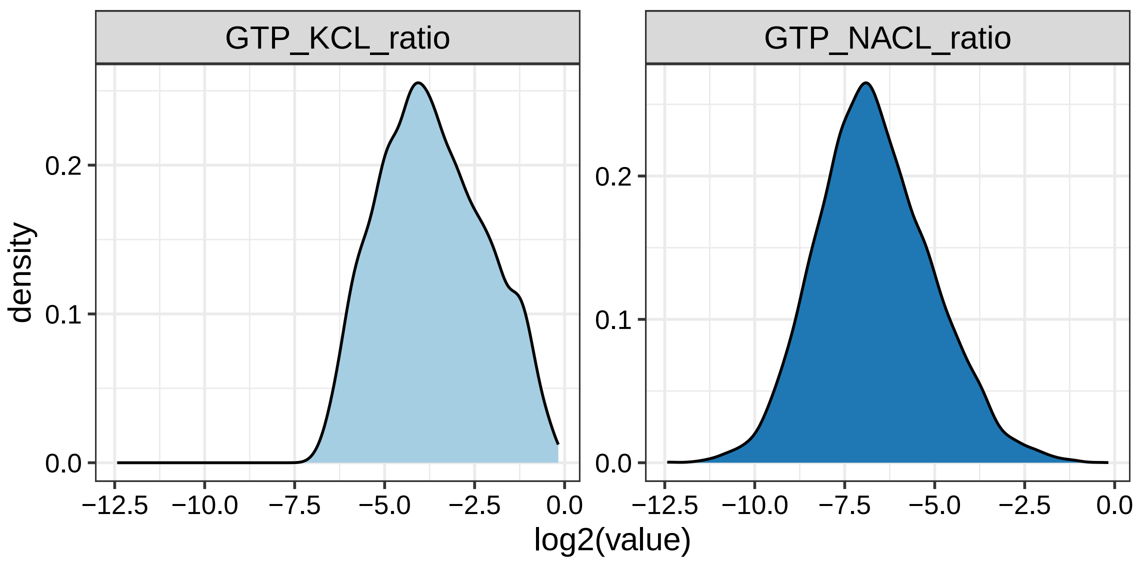

Figure 16: **Density of ratios.** G4 Peaks only.

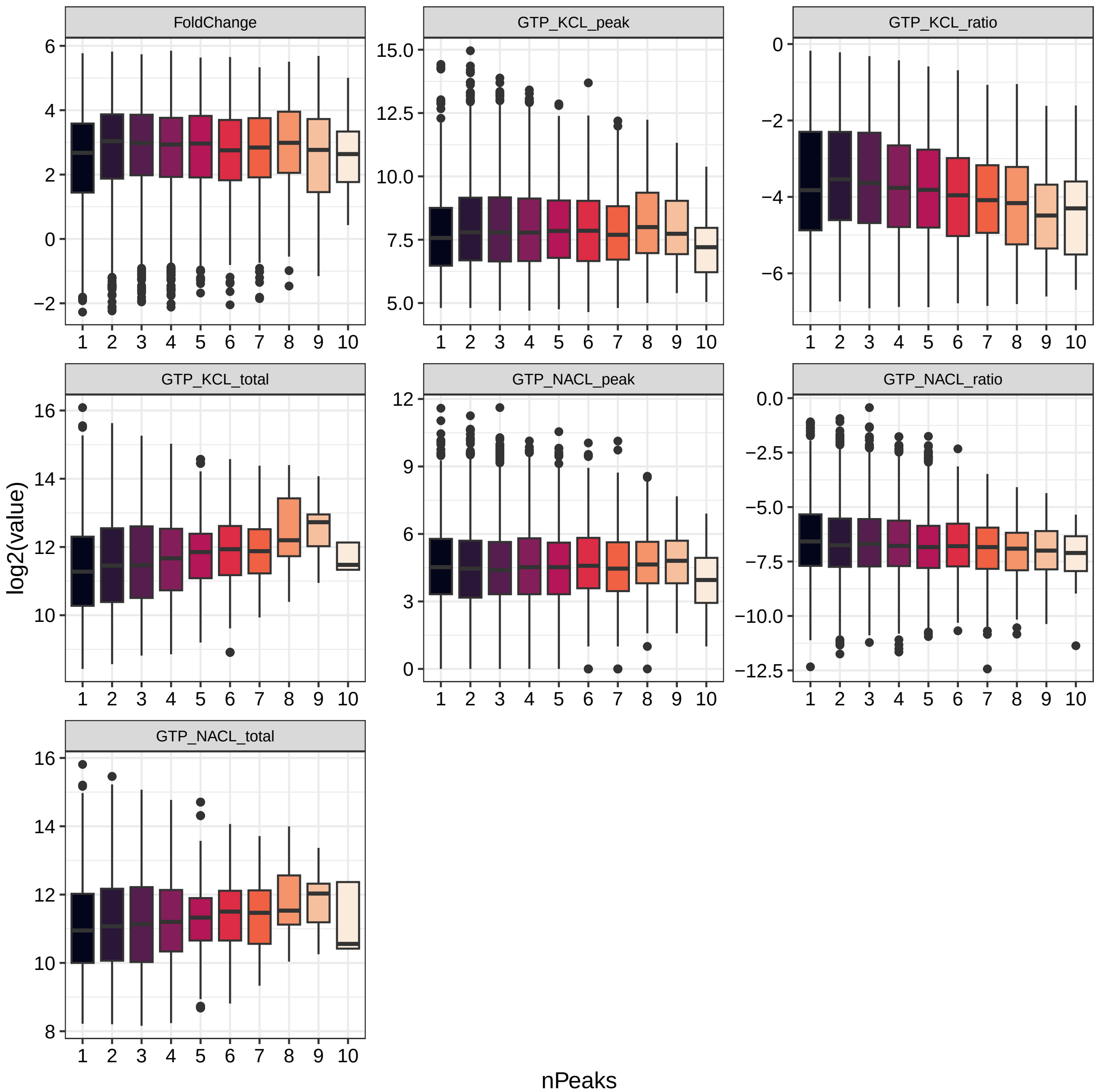

Figure 17: **Features split by number of peaks.** G4 Peaks only.

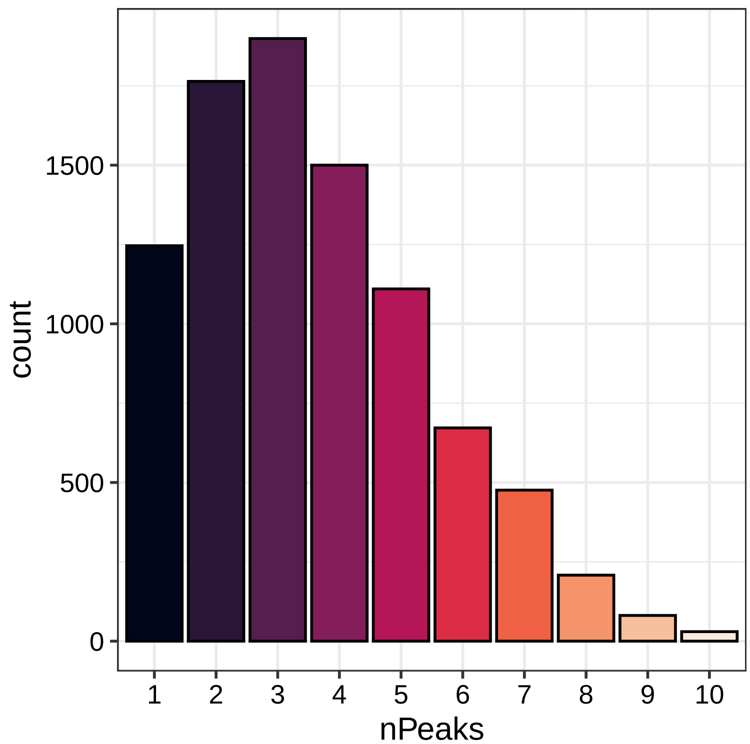

Figure 18: **Number of peaks per construct.** G4 Peaks only.

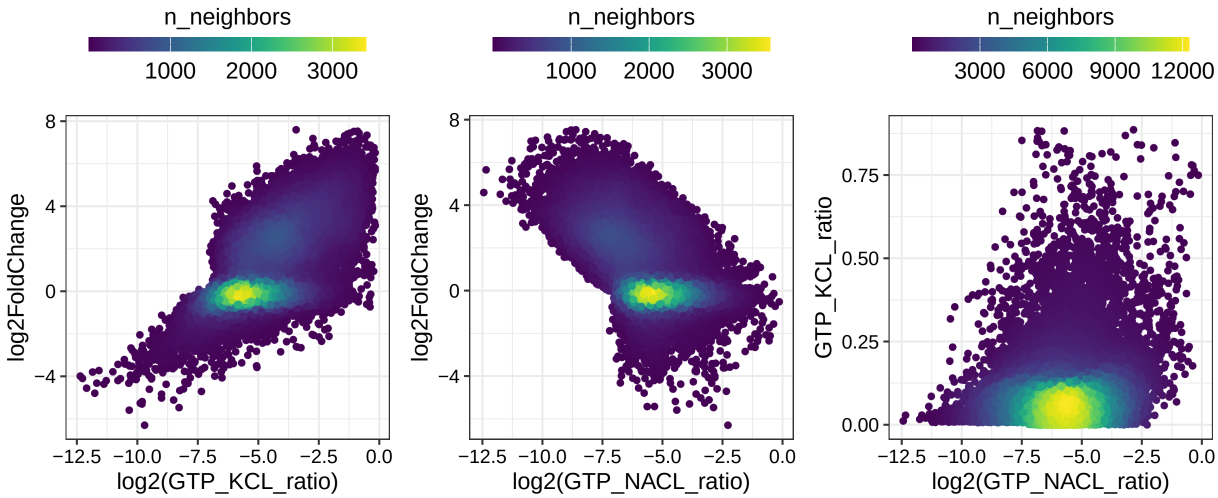

Figure 19: **Correlation between KCl/ NaCl peak ratios and the LFC.** All peaks on G4 containing constructs.

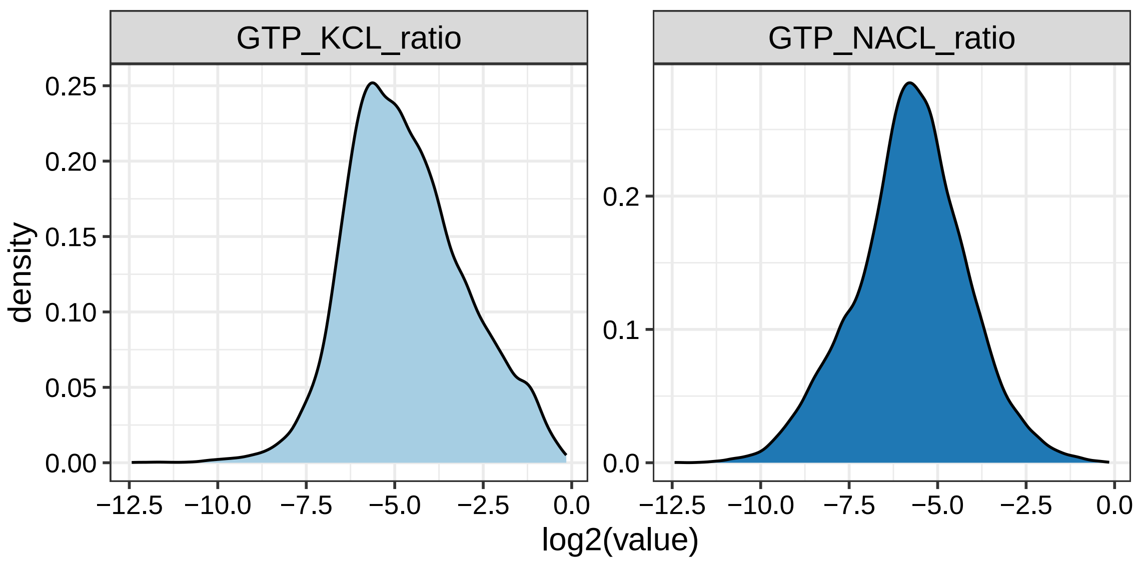

Figure 20: **Density of ratios.** All peaks on G4 containing constructs.

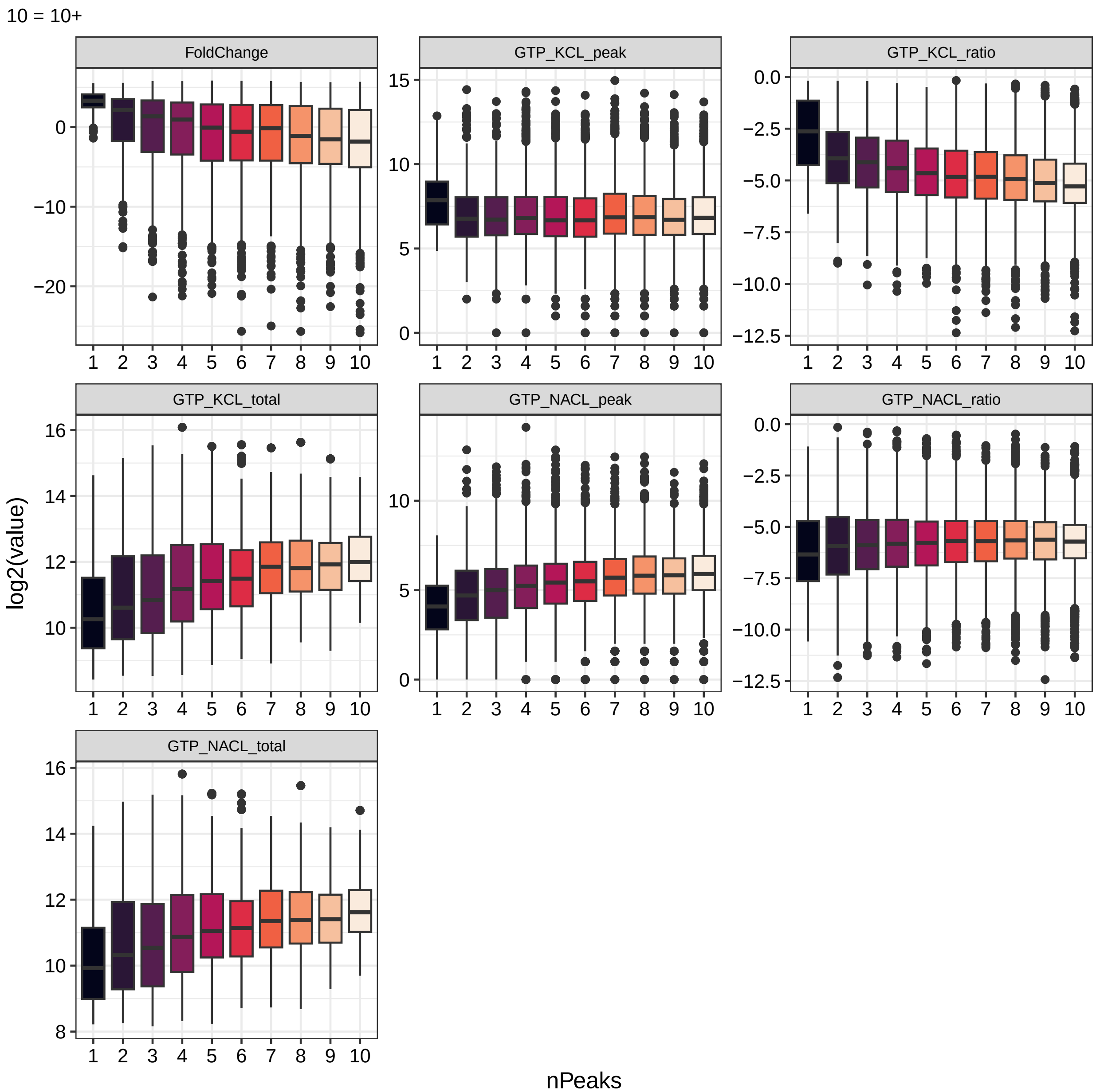

Figure 21: **Features split by number of peaks.** All peaks on G4 containing constructs.

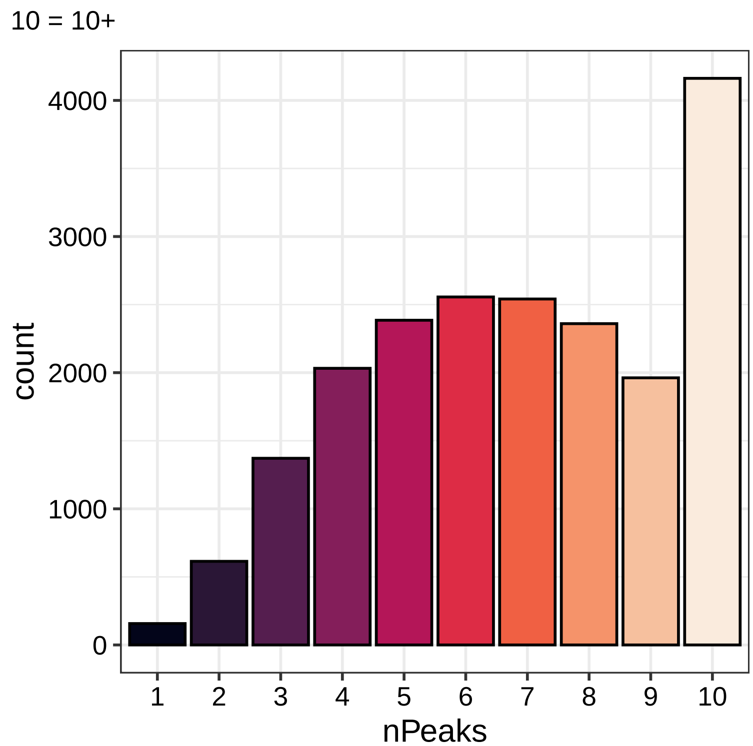

Figure 22: **Number of peaks per construct** All peaks on G4 containing constructs.

8 Sequence composition

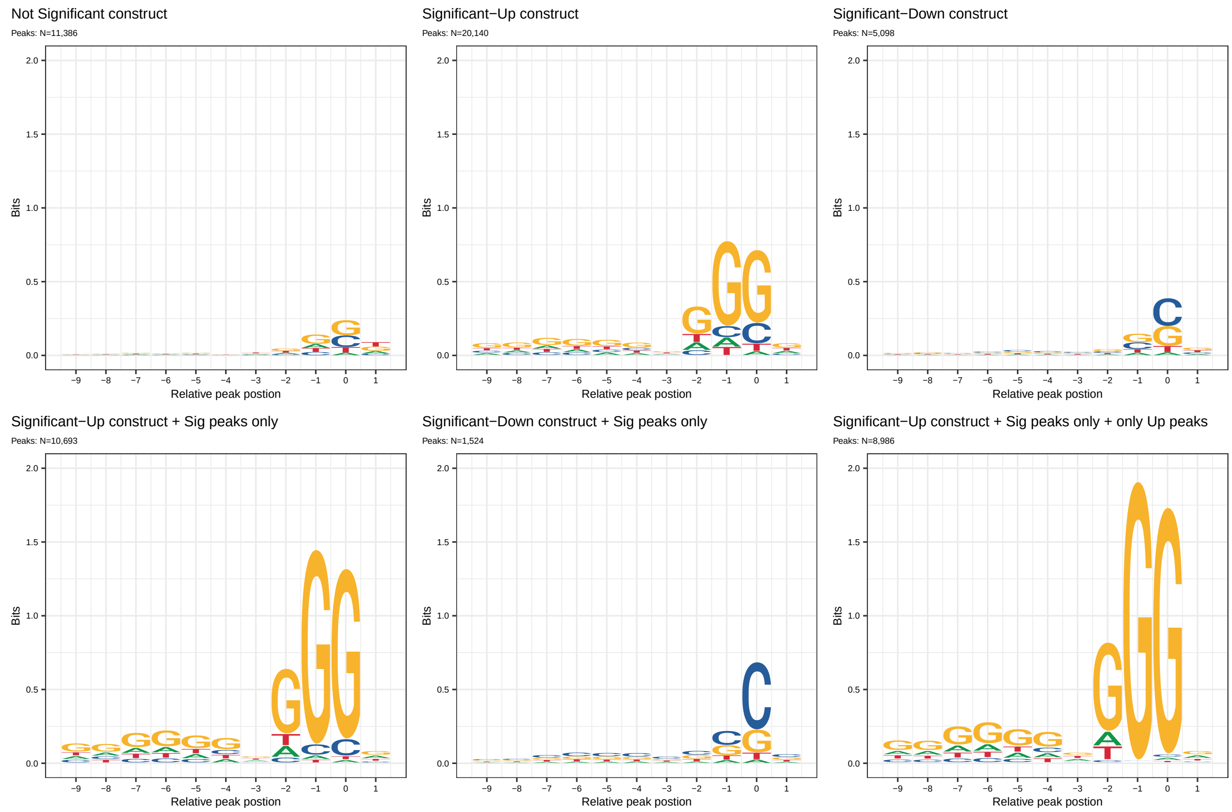

Figure 23: **G4 motif on different sets of constructs.**

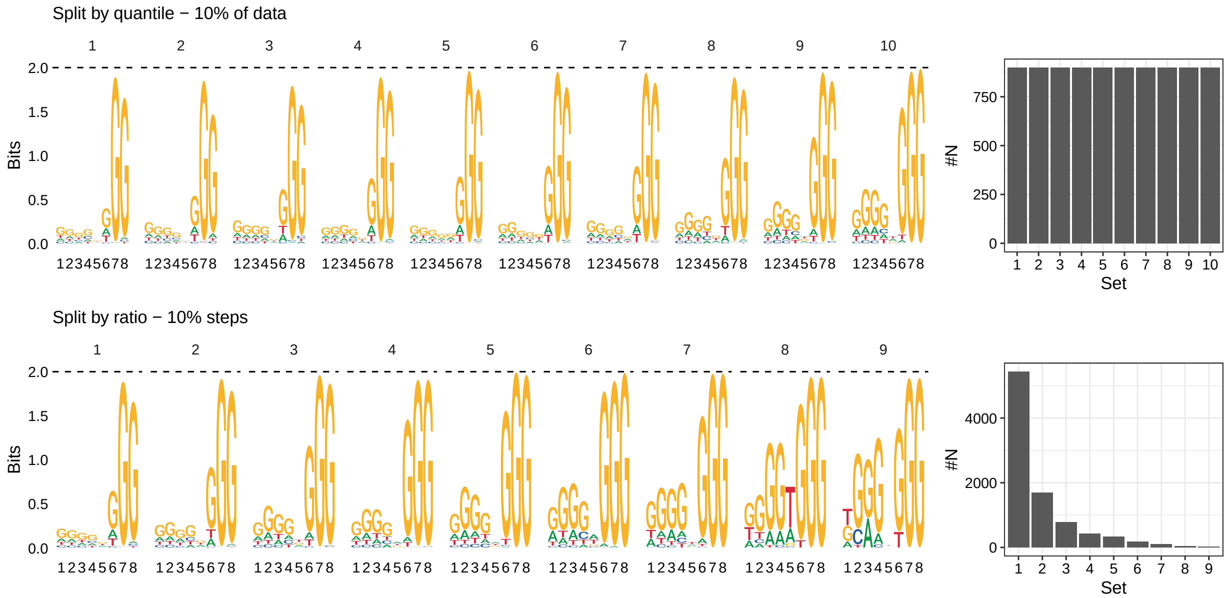

Figure 24: **G4 Peaks split by KCL ratio.**

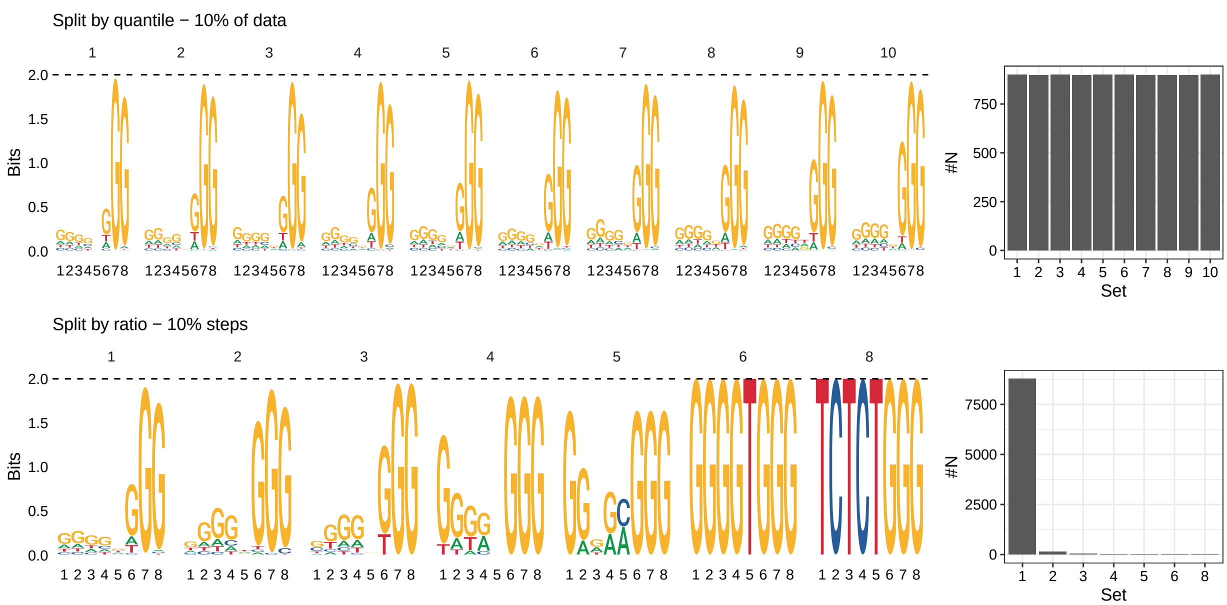

Figure 25: **G4 Peaks split by NaCl ratio.**

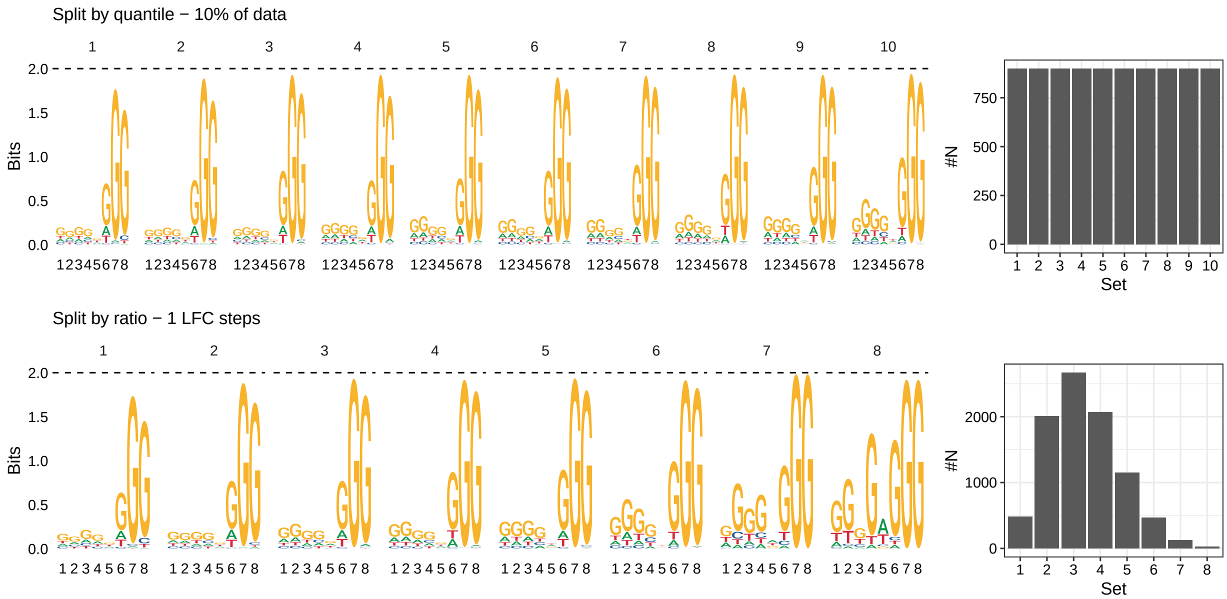

Figure 26: **G4 Peaks split by LFC.**

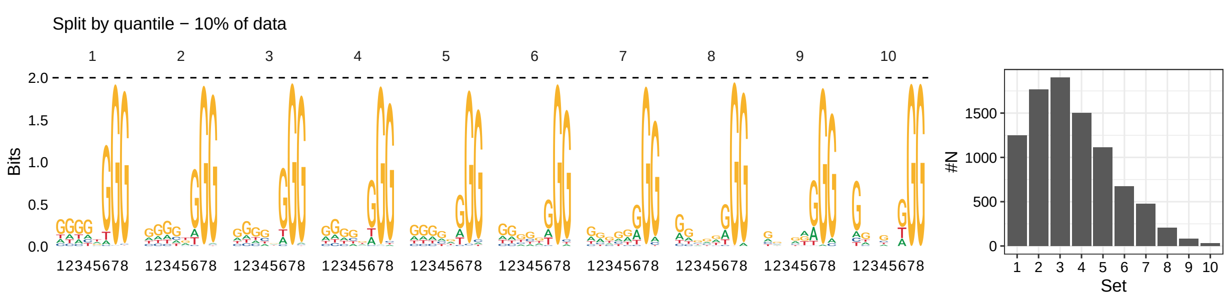

Figure 27: **G4 Peaks split by number of peaks in contstruct.**

9 Reproducibility among replicates

Read counts per replicate are comparable for each condition. This is a supplementary plot that could be used earlier in the analysis to show the reproducibility among replicates.

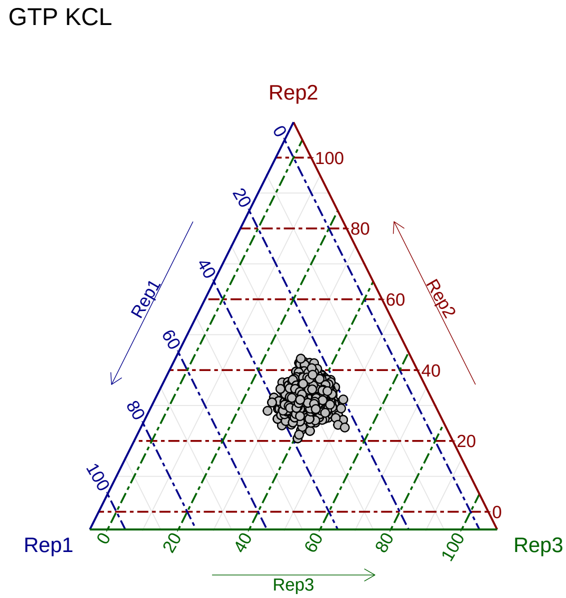

Figure 28: **Fraction of read counts per replicate**

10 Session Information

```r
sessionInfo()
```

```
## R version 4.4.2 (2024-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Sequoia 15.5
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] grid      stats4    stats     graphics  grDevices utils     datasets
## [8] methods   base
##
## other attached packages:
##  [1] ggtern_3.5.0                ggseqlogo_0.2
##  [3] xlsx_0.6.5                  BindingSiteFinder_2.4.0
##  [5] ggnewscale_0.5.0            ggrastr_1.0.2
##  [7] edgeR_4.4.2                 limma_3.62.2
##  [9] DESeq2_1.46.0               SummarizedExperiment_1.36.0
## [11] MatrixGenerics_1.18.1       matrixStats_1.5.0
## [13] ggsci_3.2.0                 tidyr_1.3.1
## [15] ggpointdensity_0.1.0        ggforce_0.4.2
## [17] patchwork_1.3.0             ggtext_0.1.2
## [19] forcats_1.0.0               ComplexHeatmap_2.22.0
## [21] viridis_0.6.5               viridisLite_0.4.2
## [23] gridExtra_2.3               ggrepel_0.9.6
## [25] knitr_1.49                  kableExtra_1.4.0
## [27] GenomicFeatures_1.58.0      UpSetR_1.4.0
## [29] reshape2_1.4.4              dplyr_1.1.4
## [31] AnnotationDbi_1.68.0        Biobase_2.66.0
## [33] ggplot2_3.5.2               rtracklayer_1.66.0
## [35] GenomicRanges_1.58.0        GenomeInfoDb_1.42.1
## [37] IRanges_2.40.1              S4Vectors_0.44.0
## [39] BiocGenerics_0.52.0
##
## loaded via a namespace (and not attached):
##   [1] bitops_1.0-9             httr_1.4.7               RColorBrewer_1.1-3
```

```
##   [4] insight_1.3.0          doParallel_1.0.17        latex2exp_0.9.6
##   [7] tools_4.4.2            backports_1.5.0          sjlabelled_1.2.0
##  [10] R6_2.5.1               DT_0.33                  ggdist_3.3.3
##  [13] GetoptLong_1.0.5       withr_3.0.2              bayesm_3.1-6
##  [16] cli_3.6.3              textshaping_0.4.1        Cairo_1.6-2
##  [19] officer_0.6.10         labeling_0.4.3           sass_0.4.9
##  [22] robustbase_0.99-4-1    SQUAREM_2021.1           readr_2.1.5
##  [25] askpass_1.2.1          mixsqp_0.3-54            Rsamtools_2.22.0
##  [28] systemfonts_1.2.0      svglite_2.1.3            invgamma_1.1
##  [31] rstudioapi_0.17.1      RSQLite_2.3.9            generics_0.1.3
##  [34] shape_1.4.6.1          BiocIO_1.16.0            car_3.1-3
##  [37] distributional_0.5.0   zip_2.3.1                Matrix_1.7-1
##  [40] ggbeeswarm_0.7.2       abind_1.4-8              lifecycle_1.0.4
##  [43] yaml_2.3.10            snakecase_0.11.1         carData_3.0-5
##  [46] SparseArray_1.6.0      blob_1.2.4               promises_1.3.2
##  [49] crayon_1.5.3           miniUI_0.1.2             lattice_0.22-6
##  [52] xlsxjars_0.6.1         KEGGREST_1.46.0          pillar_1.10.1
##  [55] rjson_0.2.23           codetools_0.2-20         glue_1.8.0
##  [58] fontLiberation_0.1.0   data.table_1.16.4        vcd_1.4-13
##  [61] vctrs_0.6.5            png_0.1-8                gtable_0.3.6
##  [64] cachem_1.1.0           xfun_0.52                openxlsx_4.2.7.1
##  [67] S4Arrays_1.6.0         mime_0.12                webr_0.1.5
##  [70] rJava_1.0-11           iterators_1.0.14         statmod_1.5.0
##  [73] devEMF_4.5-1           nlme_3.1-166             bit64_4.6.0-1
##  [76] fontquiver_0.2.1       tensorA_0.36.2.1         bslib_0.8.0
##  [79] irlba_2.3.5.1          vipor_0.4.7              colorspace_2.1-1
##  [82] DBI_1.2.3              mnormt_2.1.1             tidyselect_1.2.1
##  [85] bit_4.5.0.1            compiler_4.4.2           curl_6.1.0
##  [88] compositions_2.0-8     flextable_0.9.9          xml2_1.3.6
##  [91] fontBitstreamVera_0.1.1 DelayedArray_0.32.0     bookdown_0.43
##  [94] scales_1.3.0           DEoptimR_1.1-3-1         hexbin_1.28.5
##  [97] psych_2.5.6            lmtest_0.9-40            stringr_1.5.1
## [100] digest_0.6.37          editData_0.1.8           rmarkdown_2.29
## [103] rio_1.2.3              XVector_0.46.0           htmltools_0.5.8.1
## [106] pkgconfig_2.0.3        fastmap_1.2.0            rlang_1.1.4
## [109] GlobalOptions_0.1.2    rvg_0.3.5                htmlwidgets_1.6.4
## [112] UCSC.utils_1.2.0       shiny_1.10.0             farver_2.1.2
## [115] jquerylib_0.1.4        zoo_1.8-14               jsonlite_1.8.9
## [118] BiocParallel_1.40.0    RCurl_1.98-1.16          magrittr_2.0.3
## [121] Formula_1.2-5          GenomeInfoDbData_1.2.13  munsell_0.5.1
## [124] Rcpp_1.0.14            proto_1.0.0              gdtools_0.4.2
## [127] stringi_1.8.4          zlibbioc_1.52.0          MASS_7.3-64
## [130] plyr_1.8.9             parallel_4.4.2           sjmisc_2.8.10
## [133] Biostrings_2.74.1      gridtext_0.1.5           hms_1.1.3
## [136] circlize_0.4.16        locfit_1.5-9.12          ggpubr_0.6.0
## [139] uuid_1.2-1             ggsignif_0.6.4           XML_3.99-0.18
## [142] evaluate_1.0.3         tzdb_0.4.0               foreach_1.5.2
## [145] tweenr_2.0.3           httpuv_1.6.15            moonBook_0.3.1
```

```
## [148] openssl_2.3.1           purrr_1.0.2             polyclip_1.10-7
## [151] clue_0.3-66             ashr_2.2-63             broom_1.0.7
## [154] xtable_1.8-4            restfulr_0.0.15         rstatix_0.7.2
## [157] later_1.4.1             ragg_1.4.0              truncnorm_1.0-9
## [160] tibble_3.2.1            memoise_2.0.1           beeswarm_0.4.0
## [163] GenomicAlignments_1.42.0 cluster_2.1.8           rrtable_0.3.0
## [166] shinyWidgets_0.9.0      ztable_0.2.3
```